Pythagorean Theorem

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lesson 3 — Pythagorean Theorem	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.1

2. Standards

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Must. Find a missing leg or the hypotenuse of a right triangle from $a^2 + b^2 = c^2$, the hypotenuse found first; an irrational answer exact and, when asked, to the nearest tenth.

Must not. Stories (6.04); the coordinate plane (6.05); radicands over 225; simplifying a radical ($\sqrt{50} = 5\sqrt{2}$ is Algebra 1 — a root's decimal is the answer); 3-D.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Notes III and boards 3–4: every leg is checked against c — a leg is always shorter than the hypotenuse
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I and board 5: one theorem however the triangle is turned — the square corner first

4. Learning Target and Essential Question

Learning target. I can find a missing side of a right triangle — the hypotenuse, or a leg.

Essential question. If you know two sides of a right triangle, how do you find the third — and how do you know which side is c?

Building on. 6.01–02 — the converse and the 3-4-5 squares; 4.05 — square roots of perfect squares; Unit 2 (MA.8.NSO.1.2) — a square root between two whole numbers.

Working toward. 6.04 — the theorem inside real situations; 6.05–06 — distance on the coordinate plane; Geometry — right-triangle trigonometry.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	10	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 6 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A right triangle has legs 5 and 12. Why is the hypotenuse 13 and not 17?	DOK 2
Without computing, which is longer: the hypotenuse when the legs are 6 and 8, or a leg when the hypotenuse is 10 and the other leg is 6? How do you know?	DOK 3
Why can a missing leg never be found by adding two squares?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A right triangle has legs of 9 cm and 12 cm. How long is the hypotenuse?	***15 cm***	21 — added the legs, not their squares [MA.8.GR.1.1: the theorem is about squares]
2	A right triangle has legs of 4 inches and 9 inches. How long is the hypotenuse, to the nearest tenth?	***$\sqrt{97}$ in ≈ 9.8 in***	13 — added the legs [MA.8.GR.1.1]; 9.9 — rounded twice, 9.849 → 9.85 → 9.9: round once, from the full value
3	A right triangle has a hypotenuse of 25 ft and one leg of 20 ft. How long is the other leg?	***15 ft***	32.0 — added the squares, longer than the hypotenuse [Notes III: a leg is shorter than c]
4	A right triangle has a hypotenuse of 41 m and one leg of 40 m. How long is the other leg?	***A — 9 m***	(B) stopped at $b^2 = 81$ and forgot the square root [Math Nation names skipping the final root]; (C) subtracted the sides, 41 − 40 [MA.8.GR.1.1: the theorem is about squares]; (D) added the squares instead of subtracting: about 57.3, longer than the hypotenuse [Notes III: for a leg, subtract]
5	Two sides of this right triangle are 5 cm and 7 cm. How long is the third side, to the nearest tenth?	***$\sqrt{74}$ cm ≈ 8.6 cm***	4.9 — subtracted, taking the 7 as the hypotenuse [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]
6	A right triangle's hypotenuse is 20 inches; its legs are 16 inches and x. Which equation finds x?	***D: $16^2 + x^2 = 20^2$, so x = 12 in***	(A) the sides added, not their squares [MA.8.GR.1.1]; (B) the hypotenuse put with the legs [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (C) squared the legs but not the hypotenuse — the 20 is a side too: 20^2 [Florida B1G-M, GR.1.1: calculation errors with the theorem]
7	A right triangle has a hypotenuse of 14 m and one leg of 9 m. How long is the other leg, to the nearest tenth?	***$\sqrt{115}$ m ≈ 10.7 m***	16.6 — added the squares, longer than the hypotenuse [Notes III]
8	A pennant is cut as a right triangle with legs of 8 inches and 11 inches. The fabric costs 6 dollars. How long is its longest side, to the nearest tenth?	***$\sqrt{185}$ in ≈ 13.6 in***	19 — added the legs [MA.8.GR.1.1]; 'cannot tell' — the longest side is the hypotenuse [Notes I]
9	Jo finds a leg of a right triangle. The hypotenuse is 13 cm and the other leg is 5 cm. She writes $\sqrt{(5^2 + 13^2)} \approx 13.9$. What went wrong? How long is the leg?	***She added the squares; a leg must be shorter than 13 cm. Subtract: the leg is 12 cm.***	'13.9 is fine' — no check against the hypotenuse [Notes III]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. $\text{Leg}^2 + \text{leg}^2 = \text{hypotenuse}^2$ stays on the board all period. Every board starts by marking the square corner and the side across from it. A set-up with the numbers already squared for boards 1–2 if the squaring is the obstacle.

ELL. **Hypotenuse**: the long side across from the square corner. **Leg**: one of the two sides that make the square corner. Say "square it" and "square root" every time, with the gesture of a square.

Enrichment. Legs 1 and 1 give $\sqrt{2}$; legs $\sqrt{2}$ and 1 give $\sqrt{3}$ — keep going (the Wheel of Theodorus): which leg lengths give $\sqrt{7}$? And why is every hypotenuse in the wheel $\sqrt{n}$?

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Jo added the squares to find a leg, and her answer is longer than the 13-cm hypotenuse — impossible. A board that finds 12 without saying why 13.9 cannot be a leg has the number and not the check.

Vocabulary. **Pythagorean Theorem**; **hypotenuse**; **square root**

Materials. Whiteboards and markers. $a^2 + b^2 = c^2$ on the board for the period (memorized by the test). Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Pythagorean theorem: find the missing leg or hypotenuse length — MTM.